Exploring Deep Learning Opportunities with MIMIC-IV and SNOMED CT

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July 21, 2025

What is MIMIC-IV?

- MIMIC-IV: A freely available ICU EHR dataset (2008–2019)
- 40,000+ unique ICU patient records
- Rich structured data: demographics, diagnoses, labs, medications, procedures
- Hosted on PhysioNet: https://physionet.org/

MIMIC-IV Data Structure

- **hosp module**: hospital-wide data (admissions, labs, meds, billing)
- **icu module**: bedside ICU data (vitals, inputs, outputs, procedures)
- **note module**: free-text notes (discharge summaries, radiology)
- Linked via `subject_id`, `hadm_id`, and `stay_id`
- All tables are CSV, anonymized per HIPAA Safe Harbor

```
(anaconda3)[bhe@hpc-logon mimiciv]$ cd 0.4/
(anaconda3)[bhe@hpc-logon 0.4]$ ls
core  hosp  icu  index.html  LICENSE.txt  SHA256SUMS.txt
(anaconda3)[bhe@hpc-logon 0.4]$ ls core/
admissions.csv.gz  index.html  patients.csv.gz  transfers.csv.gz
(anaconda3)[bhe@hpc-logon 0.4]$ ls hosp/
d_hcpcs.csv.gz        d_icd_diagnoses.csv.gz   d_labitems.csv.gz  emar.csv.gz        hcpcsevents.csv.gz  labevents.csv.gz          pharmacy.csv.gz  poe_detail.csv.gz     procedures_icd.csv.gz
diagnoses_icd.csv.gz  d_icd_procedures.csv.gz  drgcodes.csv.gz    emar_detail.csv.gz  index.html          microbiologyevents.csv.gz poe.csv.gz       prescriptions.csv.gz  services.csv.gz
(anaconda3)[bhe@hpc-logon 0.4]$ ls icu/
chartevents.csv.gz  datetimeevents.csv.gz  d_items.csv.gz  icustays.csv.gz  index.html  inputevents.csv.gz  outputevents.csv.gz  procedureevents.csv.gz
```

Patient Example from MIMIC-IV (hadm_id: 28503629)

- ICU admission $\rightarrow$ ward $\rightarrow$ surgery 4$\rightarrow$ ICU readmission $\rightarrow$ discharge
- **Vitals:** High-frequency temperature data under targeted temp management
- **Procedures:** Recorded via POE, ICU system, and billing codes
- **Labs:** CRP, WBC, potassium trends across hospital stay
- **Medications:** Heparin shown via EMAR (administration timeline)

MIMIC-IV Exploration

- https://colab.research.google.com/drive/1REu-ofzNzqsTT1cxLHIegPB0nGmwKaM0?usp=sharingscrollTo=AL7Eg65fgaEj
- https://www.nature.com/articles/s41597-022-01899-x

MIMIC (Medical Information Mart for Intensive Care) is a collection of large, freely accessible clinical databases developed by the MIT Laboratory for Computational Physiology in collaboration with Beth Israel Deaconess Medical Center.

Key Features:

- Real-world, de-identified, multimodal clinical data
- Includes structured EHR, clinical notes, medical imaging, waveforms, EEG
- Covers ICU, hospital, emergency, and pediatric populations
- Widely used in ML, NLP, image analysis, and clinical outcome prediction

Access Requirements:

- Completion of CITI ”Data or Specimens Only Research” training
- PhysioNet credentialed user account

Dataset	Description
MIMIC-IV	Core dataset (v2.2): Structured records of hospital admissions, ICU stays, and emergency visits
MIMIC-IV-Note	Free-text clinical notes: discharge summaries, radiology reports, echocardiograms ,
MIMIC-CXR	Chest X-ray images with reports and CheXpert labels (over 370k images)
MIMIC-EEG	EEG signals and clinical reports for neurological research
MIMIC-Waveform	High-frequency physiological signals (ECG, BP, etc.) aligned with ICU records
MIMIC-PIC / PED	Pediatric ICU data from CHLA (currently limited access)

Access via `https://physionet.org/about/mimic`

Deep Learning on MIMIC-IV: MRSA Case Study

- Source: Nigo et al., 2024 (Nature Communications)
- Task: Predict MRSA-positive culture within 2 weeks
- Model: PyTorch_EHR using GRU-based RNN
- Dataset: Trained on MHHS, validated on MIMIC-IV (393k+ events)

Model Performance

- AUROC (MIMIC-IV):
  - PyTorch_EHR: **0.859**
  - LGBM: 0.838
  - Logistic Regression: 0.816
- PyTorch_EHR effective in low prevalence data and across event repetitions

Clinical Impact

- Reduced unnecessary MRSA antimicrobials (e.g. vancomycin)
- Early identification of high-risk patients
- High NPV in low-risk group: $>$99.8% (MIMIC-IV)
- Potential for real-time intervention and resource optimization

Limitations of ICD Coding

- Lacks semantic hierarchy and concept generalization
- Cannot distinguish similarities across clinically linked terms
- Example: MRSA bacteremia vs MRSA skin abscess = separate codes, no relation

Introducing SNOMED CT

- Comprehensive clinical terminology system
- Hierarchical relationships: ”is-a”, ”associated with”, etc.
- Synonyms, cross-references, multilingual support
- Over 300,000 concepts for diseases, findings, procedures

Enhancing Deep Learning with SNOMED CT

- Map ICD codes to SNOMED CT concepts
- Use ”is-a” and other relations to construct a concept graph
- Apply Graph Neural Networks or ontology-aware embeddings
- Enables semantic generalization across similar diagnoses

Future Research Directions

- SNOMED-enhanced GNN for infection risk prediction
- Patient-concept graphs for personalized treatment modeling
- Text + Graph multimodal models for clinical notes
- Explainable AI using SNOMED-based pathways